

# TUTORIUM-9

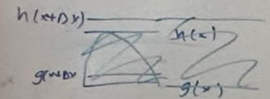
P9.2:  $f \in C(\mathbb{R}^2, \mathbb{R})$ ,  $g, h: \mathbb{R} \rightarrow \mathbb{R}$  s.d.

zz:  $F(x) := \int_{g(x)}^{h(x)} f(x, t) dt$  s.d. mit

$$F'(x) = f(x, h(x)) \cdot h'(x) - f(x, g(x)) \cdot g'(x) + \int_{g(x)}^{h(x)} \partial_x f(x, t) dt$$

$$F'(x) = \frac{d}{dx} \int_{g(x)}^{h(x)} f(x, t) dt = \lim_{\Delta x \rightarrow 0} \frac{F(x+\Delta x) - F(x)}{\Delta x}$$

$$F(x+\Delta x) - F(x) = \int_{g(x+\Delta x)}^{h(x+\Delta x)} f(x+\Delta x, t) dt - \int_{g(x)}^{h(x)} f(x, t) dt = \underbrace{\int_{g(x)}^{h(x)} (f(x+\Delta x, t) - f(x, t)) dt}_{(1)} + \underbrace{\int_{h(x)}^{h(x+\Delta x)} f(x+\Delta x, t) dt}_{(2)} - \underbrace{\int_{g(x)}^{g(x+\Delta x)} f(x+\Delta x, t) dt}_{(3)}$$



(1):  $\lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \int_{g(x)}^{h(x)} (f(x+\Delta x, t) - f(x, t)) dt$

(2):  $\lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \int_{h(x)}^{h(x+\Delta x)} f(x+\Delta x, t) dt$

$$h(x+\Delta x) = h(x) + h'(x) \Delta x + o(\Delta x)$$

MWS I.  $\exists \xi \in [h(x), h(x+\Delta x)] : \int_{h(x)}^{h(x+\Delta x)} f(x+\Delta x, t) dt = f(x+\Delta x, \xi) \cdot (h(x+\Delta x) - h(x))$

$$\Delta x \rightarrow 0 \quad \xi \rightarrow h(x)$$

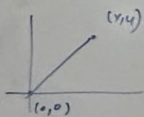
$$\lim_{\Delta x \rightarrow 0} \left[ \frac{1}{\Delta x} \underbrace{f(x+\Delta x, \xi)}_{f(x, h(x))} \cdot (h(x+\Delta x) - h(x)) \right] = f(x, h(x)) \cdot h'(x)$$

(3)  $- f(x, g(x)) \cdot g'(x)$

P9.3:  $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $F(x,y) = \begin{bmatrix} y^2 - 2x \cos y \\ 2xy + (x^2 - 1) \sin y \end{bmatrix}$

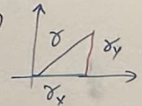
(a)  $\partial_y F_1 = \partial_x F_2 \iff \gamma_F(x,y) \text{ symm.}$

$\gamma_F(x,y) = \begin{bmatrix} \sim & 2y + 2x \sin y \\ 2y + 2x \sin y & \sim \end{bmatrix} \checkmark \text{ symm.}$



$\gamma(t) = \begin{bmatrix} x \\ y \end{bmatrix} t$   
 $[0,1] \ni t$

$$\begin{aligned} \oint (x,y) &= \int_{\gamma} F(r) \cdot dr = \int_0^1 \langle F(\gamma(t)), \dot{\gamma}(t) \rangle dt \\ &= \int_0^1 \left\langle \begin{pmatrix} t^2 y^2 - 2tx \cos(ty) \\ 2t^2 xy + (t^2 x^2 - 1) \sin(ty) \end{pmatrix}, \begin{pmatrix} x \\ y \end{pmatrix} \right\rangle dt \\ &= (\dots) \end{aligned}$$



$\gamma_x(t) = \begin{bmatrix} t \\ 0 \end{bmatrix}, t \in [0,x]$   
 $\gamma_y(t) = \begin{bmatrix} x \\ t \end{bmatrix}, t \in [0,y]$

$$\begin{aligned} \Phi(x,y) &= \int_{\gamma_y \circ \gamma_x} F(r) \cdot dr = \int_0^x F_1(t,0) dt + \int_0^y F_2(x,t) dt \\ &= \int_0^x (-2t) dt + \int_0^y (2xt + (x^2 - 1) \sin t) dt \\ &= -x^2 + xy^2 - (x^2 - 1) \cos t \Big|_0^y \\ &= -x^2 + xy^2 - (x^2 - 1) \cos y + (x^2 - 1) \\ &= xy^2 - (x^2 - 1) \cos y - 1 \end{aligned}$$

(c)  $F(x,y) = \begin{bmatrix} \partial_x \varphi \\ \partial_y \varphi \end{bmatrix} = \nabla \varphi = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$   
 $\frac{\partial \varphi}{\partial x} = F_1 = y^2 - 2x \cos y$

$\varphi = \int (y^2 - 2x \cos y) dx$   
 ①  $= xy^2 - x^2 \cos y + g(y)$   
 $\frac{\partial \varphi}{\partial y} = F_2 = 2xy + (x^2 - 1) \sin y$   
 $\varphi = \int (2xy + (x^2 - 1) \sin y) dy$   
 ②  $= xy^2 - (x^2 - 1) \cos y + h(x)$   
 $g(y) = \cos y, h(x) = 0$

→ Einstein'sche Summenkonvention:

Über doppelt auftretender Indizes innerhalb eines  
wird summiert.

$$A \underset{\substack{\psi \\ \mathbb{R}^n}}{X} \quad (Ax)_i = \sum_j A_{ij} x_j$$

• Skalarprod.  $\langle X, \psi \rangle = x_i \psi_i$

•  $X \in \mathbb{R}^n$ ,  $X = x_i e_i$

$$\operatorname{div} v = \nabla \cdot v = \sum \partial_i v_i = \partial_i v_i$$

Pg. 1:  $\nabla \times (\nabla \times v) = \nabla(\nabla \cdot v) - \Delta v$   
 $= \operatorname{grad} \operatorname{div} v - \operatorname{div} \operatorname{grad} v$

$$(a \times b)_i = \varepsilon_{ijk} a_j b_k$$

$$[\nabla \times (\nabla \times v)]_i = \varepsilon_{ijk} \partial_j (\nabla \times v)_k = \varepsilon_{ijk} \partial_j \varepsilon_{k\ell m} \partial_\ell v_m$$

$$= (\underbrace{\delta_{il} \delta_{jm}}_{i=l, j=m} - \underbrace{\delta_{im} \delta_{jl}}_{i=m, j=l}) \partial_j \partial_\ell v_m$$

$$= \delta_{il} \delta_{jm} \partial_j \partial_\ell v_m - \delta_{im} \delta_{jl} \partial_j \partial_\ell v_m$$

$$= \partial_j \partial_i v_j - \partial_j \partial_j v_i$$

$$= \partial_i \underbrace{\partial_j v_j}_{\nabla \cdot v} - (\Delta v)_i = (\nabla(\nabla \cdot v) - \Delta v)_i$$

$$\Delta = \partial_i \partial_i = \sum_i \partial_i^2$$

$\nabla \cdot \nabla = \nabla^2$   ~~$X$~~   ~~$X_i$~~

$$\delta_{ij} = \begin{cases} 1, & i=j \\ 0, & \text{sonst.} \end{cases}$$

$$\varepsilon = \begin{cases} 1, & \text{---} \\ 0, & \text{---} \\ -1, & \text{---} \end{cases}$$



$$A(x, y, z) = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad B(x, y, z) = \begin{pmatrix} x e^{-x^2-y^2} \\ y e^{-x^2-y^2} \\ 1 \end{pmatrix}$$

$$\varphi_A(x, y, z) = \frac{1}{2} (x^2 + y^2 + z^2) \Rightarrow A = \nabla \varphi \Rightarrow A \text{ Grad.feld}$$

$$\nabla \times B(x, y, z) = \begin{pmatrix} \partial_y B_z - \partial_z B_y \\ \partial_z B_x - \partial_x B_z \\ \partial_x B_y - \partial_y B_x \end{pmatrix} = \begin{pmatrix} 0 - 0 \\ 0 - 0 \\ -2xy e^{-x^2-y^2} - (-2xy) e^{-x^2-y^2} \end{pmatrix}$$

$\mathbb{R}^3$  sternförmig!

$\hookrightarrow B$  Grad.feld.

$$\begin{aligned} (1) \nabla \cdot (A \times B) &= (\nabla \times A) \cdot B - A \cdot (\nabla \times B) \\ &= 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} (1) \nabla \cdot (A \times B) &= \partial_i (A \times B)_i = \partial_i (\varepsilon_{ijk} A_j B_k) \\ &\stackrel{P.R.}{=} \varepsilon_{ijk} ((\partial_i A_j) B_k + A_j (\partial_i B_k)) \end{aligned}$$

$$\varepsilon_{ijk} = \varepsilon_{kij} = -\varepsilon_{jik}$$

$$\begin{aligned} &= (\varepsilon_{ijk} \partial_i A_j) B_k + \varepsilon_{ijk} A_j \partial_i B_k \\ &= (\varepsilon_{kij} \partial_i A_j) B_k + A_j (-\varepsilon_{jik} \partial_i B_k) \\ &= (\nabla \times A)_k B_k - A_j (\nabla \times B)_j \\ &= (\nabla \times A) \cdot B - A \cdot (\nabla \times B) \quad \square \end{aligned}$$